



Entropy-based pivotal statistics for multi-sample problems in planar shape

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Abstract

Morphometric data come from several natural and man-made phenomena; e.g., biological processes and medical image processing. The analysis of these data—known as statistical shape analysis (SSA)—requires tailored methods because the majority of multivariate techniques are for the Euclidean space. An important branch at the SSA consists in using landmark data in two dimensions, called planar shape. Hypothesis tests in the last context have been proposed to check difference of mean shapes in multiple samples. A common assumption in these tests is that involved samples have the same covariance matrix, but few works are devoted to assess the former hypothesis. This paper addresses the proposal of new entropy-based multi-samples tests for variability. To formulate these tests, we assume that pre-shape data are well described by the complex Bingham ($\mathbb{CB}$) distribution. We derive expressions for the Rényi and Shannon entropies for the $\mathbb{CB}$ and complex Watson models. Moreover, these quantities are understood as entropy-based tests to assess if multiple spherical samples have the same degree of disorder, a kind of variability. To quantify the performance of proposed tests, we perform a Monte Carlo study, adopting empirical test sizes and powers as comparison criteria. An application to real data using the second thoracic vertebra T2 of mice is done to assess possible effects of body weight on the shape of mouse vertebra. Numerical results indicate that the test based on the optimized Rényi entropy may consist in a good tool to recognize variability similarity on planar-shape data.

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1 Introduction

Statistical shape analysis (SSA) has been receiving great attention since (Kendall 1984; Bookstein 1978, 1984). The pattern recognition of biological specimens is a good example of the SSA tools utilization to solve real problems. The main concept of SSA is the meaning of shape, which is defined how what remains after removing location, scale and rotation. In mathematical terms, shapes are multidimensional points over the unit sphere. Applying multivariate analysis tools to these data often fails because they are not appropriate for non-Euclidean spaces. This last fact illustrates the relevance of using SSA.

A crucial step in SSA is to define a probability model to describe planar shape data. One of the most successful distributions in planar shape is the Bingham model. Its real version was pioneered by Bingham (1974) as a generalization of the Dimroth–Watson law (Watson 1965) and its complex counterpart as well as some of its mathematical properties was presented by Kent (1994). In this paper, we assume the complex Bingham ($\mathbb{CB}$) model as probabilistic assumption. This paper provides new planar-shape tools to check commonly variability in spherical samples based on information theory measures. As a relevant contribution to numerical evaluations of the $\mathbb{CB}$ distribution, Kent et al. (2004) have proposed one of its random number generators based on the truncated multivariate exponential law. Dore et al. (2016) have proposed a bias-corrected estimation method for its $\mathbb{CB}$ model parameters. Amaral et al. (2007) have derived a novel hypothesis test based on the lambda statistics and bootstrap resampling methods. In additional, Amaral et al. (2010) have furnished bootstrap procedures for constructing confidence regions for the mean shape of objects labeled by landmarks in two dimensions. Amaral et al. (2013) have provided methods to detect influential observations in a sample of pre-shapes under such model. On the other hand, Dryden et al. (2005) have proposed a mathematical treatment to the statistical analysis of high-dimensional spherical data and shape spaces. Approximations in the maximum likelihood (ML)-based inference context were considered for several models, such as $\mathbb{CB}$ and complex Watson ($\mathbb{CW}$) distributions.

According to Kullback (1997), information theory has its mathematical fundamentals in the concept of disorder or entropy in thermodynamics and statistical mechanics. The entropy can be interpreted not only as the uncertainty measure as also an information measure (Rényi 1961). In this paper, the concept of entropy is understood as a variability multivariate measure, such as discussed by Chen et al. (2016). In recent years, entropy classes have been used in statistical inference for various purposes; e.g., studying asymptotic properties of estimators alternative to the ML method (Toma and Broniatowski 2011) and proposing robust hypothesis tests (Gul and Zoubir 2016). Lin (1991) has introduced a new class of information-theoretic divergence measures based on the Shannon entropy. He determined the relationship of such measures with the variational distance, and the probability of misclassification error was established in

terms of bounds. Nadarajah and Zografos (2005) have derived exact forms of Rényi and Shannon entropies for various continuous bivariate distributions. Contreras-Reyes (2015) has developed exact expressions of Rényi entropy for skew-normal distributions and related families. Pietrzak et al. (2016) have proposed a frequency-based framework to obtain the effective number of species (named the Hill numbers) using empirical Rényi entropy and divergence. Chen et al. (2016) have provided new insights about entropy behaviors as a multivariate variability measure. Seo and Kang (2014) have furnished the entropy of a generalized half-logistic distribution based on Type-II censored samples from Bayesian perspectives. Song et al. (2017) have developed two estimation methods based on the principles of maximum and ordinary entropies for the four-parameter exponential gamma distribution. More information on these stochastic measures can be found in Rényi (1961), Kailath (1967), Diaconis and Zabell (1982), MacKay (2003) and Pardo (2005).

In this paper, we focus on hypothesis tests based on information theory measures for multi-samples in planar-shape data. To obtain the associated pivotal statistics, we derive the Shannon and Rényi entropies for the $\mathbb{CB}$ and $\mathbb{CW}$ models. Here, we use the entropy measure as a way of quantifying the variability in point vectors over the complex sphere. Thus, two hypothesis tests are provided. We make a Monte Carlo simulation study to quantify the performance of our proposals, adopting empirical test size and power as figures of merit. Results point out hypothesis tests based on the “optimized” Rényi entropy present the best performance under both spherical high and low concentrations. Finally, an application to real data is made in the same context of Dryden and Mardia (2016) and Kent (1994). Numerical evidence favors the Rényi entropy test having its optimized order parameter.

The paper unfolds as follows. Section 2 presents a brief discussion on the $\mathbb{CB}$ distribution. Section 3 tackles the hypothesis tests for the equality of the entropies due to r independent populations based on information theory measures. Numerical results are displayed in Sect. 4. The conclusion remarks are presented in Sect. 5. Finally, proofs of the theoretical results are given in Appendix A.

2 Some concepts about shapes and properties of the complex Bingham distribution

The main concept of statistical shape analysis (SSA) is the meaning of shape (Kent 1995). One way to describe a shape is to indicate a finite set of points in the boundary of the object. This finite number of points of each object is called *landmarks*. In what follows, a brief overview is given of concepts necessary for the understanding of the SSA. The mathematical representation of landmarks in an object is called configuration.

A configuration matrix X is a $k \times m$ matrix of Cartesian coordinates of the $k + 1$ landmarks in m dimensions. A configuration space is a space of all possible landmark coordinates. In this paper, we consider $k \geq 2$ landmarks number in $m = 2$ dimensions, called of *planar shape*. Thus, the configuration matrix (Dryden and Mardia 2016) is given by

$$\mathbf{X} = [x_1 \ x_2]^\top = \begin{pmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \\ \vdots & \vdots \\ x_{k+1,1} & x_{k+1,2} \end{pmatrix}, \quad (1)$$

where $x_{j,1}$ and $x_{j,2}$ indicate coordinates in real and imaginary axes, respectively. Some transformations must be made in $\mathbf{X}$ to remove the effects of location, scale and rotation. For $m = 2$, the mathematical configuration must be rewritten as a complex vector. Define a $(k + 1) \times 1$ complex vector

$$\mathbf{z}^0 = [z_1^0, z_2^0, \dots, z_{(k+1)}^0]^\top = [x_{1,1} + ix_{1,2}, \dots, x_{(k+1),1} + ix_{(k+1),2}]^\top \text{ for } i = \sqrt{-1}, \quad (2)$$

which corresponds to the complex coordinates for landmarks. To describe the shape of an object, it is necessary to specify a suitable coordinate system; i.e., a system that is invariant under translation, scaling and rotation of the configuration. In this paper, we will use the *Kendall coordinate* systems (Kendall 1984). To get these coordinates, we need to define the Helmert submatrix, the j th row of the Helmert submatrix $\mathbf{H}$ is given by

$$(h_j, \dots, h_j, -jh_j, 0, \dots, 0) \text{ for } h_j = -[j(j+1)]^{-1/2}.$$

So the j th row of $\mathbf{H}$ consists of h_j repeated j times, followed by $-jh_j$ and $k - j$ zeros for $j = 1, \dots, k$. To remove the location effect in $\mathbf{z}^0 = [z_1^0, \dots, z_{(k+1)}^0]^\top$, we pre-multiply it by the Helmert submatrix $\mathbf{H}$; i.e.,

$$\mathbf{z}_H = \mathbf{H}\mathbf{z}^0.$$

The origin is removed because coincident landmarks are not allowed. We refer to $\mathbf{z}_H$ as the *Helmertized landmark* coordinates. To remove the scale effect, we divided the obtained Helmertized configuration $(\mathbf{z}_H)$ by its norm

$$\mathbf{z} = \frac{\mathbf{z}_H}{\|\mathbf{z}_H\|} = \frac{\mathbf{z}_H}{\sqrt{\mathbf{z}_H^* \mathbf{z}_H}},$$

where $(\cdot)^*$ represents the complex conjugate operator and $\|\cdot\|$ denotes the norm of a complex vector. According to Kendall (1984), this normalized vector $\mathbf{z}$ is called *pre-shape*, once the shape is an equivalence class under the action of the group of similarity transformations. In order to visualize shapes, it is often convenient to choose a particular member of the shape, which is taken as being representative of the shape denominated by *icon*. An important step in SSA (in order to detect changes and variability in shape) is to define a probability model to describe planar-shape data. One of pre-shape main distributions is the $\mathbb{C}B$ model (Dryden and Mardia 2016). We briefly discuss the $\mathbb{C}B$ distribution, admitting each k -dimensional random vector is associated to an object in the plan with $k + 1$ landmarks. The $\mathbb{C}B$ distribution is defined on the unit complex sphere in $\mathbb{C}^k$ for $k \geq 2$, say $\mathbb{C}S^{k-1} = \{\mathbf{z} = (z_1, \dots, z_k)^\top : \mathbf{z}^* \mathbf{z} = 1\}$, and its probability density function (pdf) is given by

$$f(\mathbf{z}; \mathbf{A}) = c(\mathbf{A})^{-1} \exp(\mathbf{z}^* \mathbf{A} \mathbf{z}), \quad \mathbf{z} \in \mathbb{C}S^{k-1}, \quad (3)$$

where $(\cdot)^*$ represents the complex conjugate of the transpose of z , $\mathbf{A}$ is a $k \times k$ Hermitian matrix (i.e., $\mathbf{A} = \mathbf{A}^*$) and $c(\mathbf{A})$ is a normalizing constant. This case is denoted as $\mathbf{z} \sim \mathbb{CB}_{k-1}(\mathbf{A})$. The normalizing constant $c(\cdot)$ has closed-form given by (Kent 1994)

$$c(\mathbf{A}) = 2\pi^k \sum_{j=1}^k a_j \exp(\lambda_j) \quad \text{with} \quad a_j^{-1} = \prod_{\substack{i=1 \\ i \neq j}}^k (\lambda_j - \lambda_i), \quad (4)$$

where λ_j for $j = 1, \dots, k$ represent the eigenvalues of $\mathbf{A}$ and $\lambda_1 < \lambda_2 < \dots < \lambda_k = 0$. The last identity is necessary to become the $\mathbb{CB}$ model identifiable (for details, see Dryden and Mardia 2016, p. 221). Further, this normalizing constant is such that $c(\mathbf{A}) = c(\mathbf{\Delta})$, where $\mathbf{\Delta} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_k)$ and $\text{diag}(\cdot)$ denotes the diagonal matrix of a vector argument. This distribution was introduced by Kent (1994) and provides a useful structure for analyzing planar-shape data. An important $\mathbb{CB}$ property is that this model is invariant under scalar rotation; i.e., $\mathbf{z}$ and $\exp(i\theta)\mathbf{z}$ have the same distribution for all $\theta \in [0, 2\pi)$. Dryden and Mardia (2016) state that if a $\mathbb{CB}$ distributed object is rotated, then its density and statistical properties are similar to those due to object at the original rotation. The $\mathbb{CB}$ model can be obtained by conditioning a zero mean multivariate complex normal ($\mathbb{CN}$) distribution to have unit norm. In particular, if $\boldsymbol{\omega} \sim \mathbb{CN}_k(0, \boldsymbol{\Sigma})$ then $\boldsymbol{\omega} \mid \{\|\boldsymbol{\omega}\| = 1\} \sim \mathbb{CB}_{k-1}(-\frac{1}{2}\boldsymbol{\Sigma}^{-1})$, where $\mathbb{CN}_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the k -variate $\mathbb{CN}$ model with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.

An important special case of the $\mathbb{CB}$ distribution is the $\mathbb{CW}$ distribution. Mardia and Dryden (1999) proved that when there are just two distinct eigenvalues in $\mathbf{A}$ (a single distinct largest eigenvalue and all other eigenvalues being equal), $\mathbf{A}$ can reparameterized as $\mathbf{A} = -\kappa(\mathbf{I}_k - \boldsymbol{\mu}\boldsymbol{\mu}^*)$ and (3) collapses in

$$f(\mathbf{z}; \boldsymbol{\mu}, \kappa) = c_1(\kappa)^{-1} \exp \left\{ \kappa \cos^2 \left[\rho(\mathbf{H}^\top \mathbf{z}, \mathbf{H}^\top \boldsymbol{\mu}) \right] \right\}, \quad \mathbf{z} \in \mathbb{CS}^{k-1},$$

where $\rho(\mathbf{H}^\top \mathbf{z}, \mathbf{H}^\top \boldsymbol{\mu}) = \cos^{-1}(|\mathbf{z}^* \boldsymbol{\mu}|)$, for $0 \leq \rho(\mathbf{H}^\top \mathbf{z}, \mathbf{H}^\top \boldsymbol{\mu}) \leq \pi/2$, $|\cdot|$ represents the norm of a complex number, κ is a concentration parameter, $\boldsymbol{\mu}$ is the modal vector on the pre-shape sphere and ρ is the Procrustes distance.¹ Note that $\mathbf{H}^\top \mathbf{z}$ is the icon corresponding to the pre-shape $\mathbf{z}$. This case is denoted as $\mathbf{z} \sim \mathbb{CW}_{k-1}(\boldsymbol{\mu}, \kappa)$.

The integrating constant defined in Mardia and Dryden (1999) is given by

$$c_1(\kappa) = \frac{2\pi^k}{(k-1)!} {}_1F_1(1; k; \kappa),$$

where

$${}_1F_1(a; b; x) = 1 + \frac{a}{b} \frac{x}{1!} + \frac{a(a+1)}{b(b+1)} \frac{x^2}{2!} + \frac{a(a+1)(a+2)}{b(b+1)(b+2)} \frac{x^3}{3!} + \dots, \quad \text{with } a, b, x \in \mathbb{R},$$

is the confluent hypergeometric function (see Abramowitz and Stegun 1964). The concentration parameter (κ) can assume negative values, but the assumption $\kappa > 0$

¹ According to Dryden and Mardia (2016), the Procrustes distance $\rho(\cdot, \cdot)$ is considered as the smallest angle (with respect to rotations of the pre-shapes) between the vectors corresponding to $\mathbf{H}^\top \mathbf{z}$ and $\mathbf{H}^\top \boldsymbol{\mu}$.

is the most common. For positive concentration, the modal pre-shape direction is $\mu \exp(i\theta)$, where $\theta \in [0, 2\pi)$ is an arbitrary rotation angle. If $\kappa = 0$, one gets the uniform distribution in the complex unitary sphere. According to Mardia and Dryden (1999) if κ is large, then the $\mathbb{C}W$ model tends to a complex multivariate normal distribution in the tangent space with mean μ and covariance matrix $(2\kappa)^{-1}(\mathbf{I}_k - \mu\mu^*)^-$, where $(x)^-$ represents the generalized inverse of x . In this paper, we consider the Moore–Penrose version. More details on the $\mathbb{C}B$ and $\mathbb{C}W$ distributions can be found in Mardia and Dryden (1999), Bingham et al. (1992) and Dryden (1991).

2.1 Maximum likelihood estimation of the $\mathbb{C}B$ distribution

Let $z_1, z_2, \dots, z_n \in \mathbb{C}S^{k-1}$ for $n \geq k$ be an observed sample drawn from $z \sim \mathbb{C}B_{k-1}(\mathbf{A})$ and

$$\mathbf{S} = \sum_{j=1}^n z_j z_j^* \quad (5)$$

be a $k \times k$ Hermitian matrix, termed by *product matrix*. Suppose that the eigenvalues of $\mathbf{S}$ are positive, distinct and denoted by $0 < l_1 < \dots < l_k$ and let $\mathbf{g}_1, \dots, \mathbf{g}_k$ be the corresponding eigenvectors. According to Dryden and Mardia (2016), the $\mathbb{C}B$ log-likelihood function at $\mathbf{A}$ is given by

$$\ell(\mathbf{A}; \mathbf{z}) = \sum_{i=1}^n z_i^* \mathbf{A} z_i - n \log c(\mathbf{A}) = \sum_{j=1}^k \lambda_j \mathbf{y}_j^* \mathbf{S} \mathbf{y}_j - n \log c(\mathbf{\Delta}),$$

where $\mathbf{\Delta} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_k)$ and λ_j are the eigenvalues of matrix $\mathbf{A}$. Considering $\lambda_1 < \lambda_2 < \dots < \lambda_{k-1} < \lambda_k = 0$ constant, the maximum likelihood estimate (MLE) of eigenvectors, say $\mathbf{y}_j$, referring to matrix $\mathbf{A}$ are given by $\hat{\mathbf{y}}_j = \mathbf{g}_j$, for $j = 1, \dots, k$. So,

$$\ell(\mathbf{A}; \mathbf{z}) = \sum_{j=1}^k l_j \lambda_j - n \log c(\mathbf{\Delta}). \quad (6)$$

Deriving the expression (6), we find the system of equations

$$\frac{\partial \ell(\mathbf{A}; \mathbf{z})}{\partial \lambda_j} = \frac{1}{n} l_j, \quad j = 1, \dots, k-1. \quad (7)$$

It is noticeable there is not closed-form solution for MLEs of eigenvalues. Thus, it is need to use interactive process. Dryden and Mardia (2016) have proved under high concentration that

$$\hat{\lambda}_j \cong -n/l_j \quad \text{for } j = 1, \dots, k-1.$$

Details about MLEs can be found in Amaral et al. (2013) and Dore et al. (2016). As additional information on the $\mathbb{C}B$ inferential context, Amaral et al. (2013) have derived the inverse of the Fisher information matrix (FIM) as:

$$I_F^{-1}(\mathbf{A}) = \left\{ -\frac{n}{[c(\mathbf{A})]^2} \left[\frac{\partial c(\mathbf{A})}{\partial \lambda_j} \frac{\partial c(\mathbf{A})}{\partial \lambda_s} - \frac{\partial^2 c(\mathbf{A})}{\partial \lambda_j \partial \lambda_s} c(\mathbf{A}) \right] \right\}^{-1}, \quad (8)$$

where

$$\frac{\partial c(\mathbf{A})}{\partial \lambda_i} = 2\pi^k \left[\sum_{\substack{j=1 \\ j \neq i}}^k \frac{a_j \exp(\lambda_j)}{(\lambda_j - \lambda_i)} + a_i \exp(\lambda_i) + a_i b_i \exp(\lambda_i) \right], \quad (9)$$

$a_j^{-1} = \prod_{\substack{l=1 \\ l \neq j}}^k (\lambda_j - \lambda_l)$ and $b_i = \sum_{l \neq i} \frac{1}{(\lambda_i - \lambda_l)}$. For the second derivative, Amaral et al. (2013) considered two cases:

$$\frac{\partial^2 c(\mathbf{A})}{\partial \lambda_j \partial \lambda_s} = \begin{cases} 2\pi^k \left\{ \sum_{\substack{l=1 \\ l \neq j, s}}^k \frac{a_l \exp(\lambda_l)}{(\lambda_l - \lambda_j)(\lambda_l - \lambda_s)} - \left[\frac{a_s b_s \exp(\lambda_s)}{(\lambda_s - \lambda_j)} + \frac{a_j b_j \exp(\lambda_j)}{(\lambda_j - \lambda_s)} \right] \right. \\ \left. + \left[\frac{a_s \exp(\lambda_s)}{(\lambda_s - \lambda_j)} + \frac{a_j \exp(\lambda_j)}{(\lambda_j - \lambda_s)} \right] - \left[\frac{a_s \exp(\lambda_s)}{(\lambda_s - \lambda_j)^2} + \frac{a_j \exp(\lambda_j)}{(\lambda_j - \lambda_s)^2} \right] \right\}, & \text{if } s \neq j \\ 2\pi^k \left\{ 2 \sum_{\substack{l=1 \\ l \neq j}}^k \frac{a_l \exp(\lambda_l)}{(\lambda_l - \lambda_j)^2} + a_j \exp(\lambda_j)(w_j + b_j^2 - 2b_j + 1) \right\}, & \text{if } s = j. \end{cases}$$

where $w_j = \sum_{l \neq j} (\lambda_j - \lambda_l)^{-1}$.

3 Test for the equality of the entropies of r independent populations in planar shape

In this section, we derive the Rényi and Shannon entropies for the $\mathbb{C}B$ and $\mathbb{C}W$ models. We present some theoretical results involving these quantities in order to obtain pivotal statistics for detecting the common variability in the r independent samples on the pre-shape unit complex sphere. This hypothesis is important because it is often assumed in tests for the mean shape. The hypothesis tests constructed by means of these pivotal statistics aim to assess the equality of entropies.

3.1 Entropy measures

Let X be a random variable taking values on a sample space $\mathcal{X}$ and having distribution defined at the parameter vector θ . Let $(\mathcal{X}, \mathcal{A}, \mathcal{P}_\theta)_{\theta \in \Theta}$ be the statistical space associated with the random variable X , where $\mathcal{A}$ is the σ -field designed from $\mathcal{X}$ and $\{\mathcal{P}_\theta\}_{\theta \in \Theta}$ indicates a family of probability distributions defined on the measurable space $(\mathcal{X}, \mathcal{A})$ and Θ is the parametric space. We assume that the probability distributions $\{\mathcal{P}_\theta\}$ are absolutely continuous with respect to a σ -finite measure μ (Lebesgue measure) on $(\mathcal{X}, \mathcal{A})$.

Shannon (1948) introduced the entropy measure

$$H_S(\boldsymbol{\theta}) = \mathbb{E}_X[-\log(f(X; \boldsymbol{\theta}))] = - \int_{\mathcal{X}} [\log(f(x; \boldsymbol{\theta}))] f(x; \boldsymbol{\theta}) d\mu(x). \quad (10)$$

Rényi (1961) proposed the following generalization of the Shannon entropy:

$$H_R^\beta(\boldsymbol{\theta}) = \frac{1}{1-\beta} \log \left[\mathbb{E}_X(f(X; \boldsymbol{\theta})^{\beta-1}) \right] = \frac{1}{1-\beta} \log \int_{\mathcal{X}} f(x; \boldsymbol{\theta})^\beta d\mu(x), \quad (11)$$

which is valid only for $\beta \in \mathbb{R}_+ - \{1\}$.

Golshani and Pasha (2010) and Contreras-Reyes (2015) have provided some relevant properties for Shannon and Rényi entropies, listed below:

1. The Rényi entropy can be negative;
2. The Rényi entropy is invariant under a location transformation of the random variable; i.e., for $m \in \mathbb{R}$,

$$H_R^\beta(\mathbf{x} + m) = H_R^\beta(\mathbf{x});$$

3. The Rényi entropy is not invariant under a scale transformation of the random variable. For example, for $m \in \mathbb{R}$,

$$H_R^\beta(m\mathbf{x}) = H_R^\beta(\mathbf{x}) + |\log(m)|;$$

4. For $\beta_1 < \beta_2$, we have $H_R^{\beta_1}(\mathbf{x}) \geq H_R^{\beta_2}(\mathbf{x})$ for all $\mathbf{x}$, and the equality holds if and only if $\mathbf{x}$ is a uniform random variable;
5. The Rényi entropy converge for Shannon entropy for all $\beta \in \mathbb{R}_+ - \{1\}$, that is

$$\lim_{\beta \rightarrow 1} H_R^\beta(\mathbf{x}) = H_S(\mathbf{x}).$$

Additional properties about Rényi and Shannon entropies can be found in Pardo (2005), Cover and Thomas (1991) and Sánchez-Moreno et al. (2014).

In this paper, we derive new entropy expressions on planar shape. Replacing X in (10) and (11) by $\mathbf{z} \sim \mathbb{CB}_{k-1}(\mathbf{A})$, the following theorem holds.

Theorem 1 *Let $\mathbf{z} \sim \mathbb{CB}_{k-1}(\mathbf{A})$. For $\beta \in \mathbb{R}_+ - \{1\}$, we have that the Rényi and Shannon entropies are given by*

$$H_R^\beta(\mathbf{A}) = \frac{1}{(1-\beta)} [\log c(\beta\mathbf{A}) - \beta \log c(\mathbf{A})] \quad (12)$$

and

$$H_S(\mathbf{A}) = \log c(\mathbf{A}) - \sum_{i=1}^{k-1} \lambda_i \frac{1}{c(\mathbf{A})} \frac{\partial c(\mathbf{A})}{\partial \lambda_i}. \quad (13)$$

As discussed in Sect. 2, the $\mathbb{C}W$ is a special case of the $\mathbb{C}B$ model having two parameters: concentration κ and modal vector $\boldsymbol{\mu}$. The former parameter is the most explored in practice and larger values of $|\kappa|$ indicate more concentrated scenes. Now, first we derive expressions for Rényi and Shannon entropies in Theorem 1. Subsequently, we present the behavior of these entropies when κ tends to zero.

Corollary 1 *Let $\mathbf{z} \sim \mathbb{C}W_{k-1}(\boldsymbol{\mu}, \kappa)$ where $\boldsymbol{\mu}$ is modal vector of the spherical pre-shape and κ is concentration parameter. Then,*

(a) *the Rényi entropy is given by*

$$H_R^\beta(\boldsymbol{\mu}, \kappa) = \frac{1}{(1 - \beta)} \left\{ -\beta \log[c_1(\kappa)] + \log[c_1(\beta\kappa)] \right\} \quad (14)$$

(b) *and the Shannon entropy is*

$$H_S(\boldsymbol{\mu}, \kappa) = \log[c_1(\kappa)] - \frac{\kappa}{k} \cdot \frac{{}_1F_1(2; k+1; \kappa)}{{}_1F_1(1; k; \kappa)}. \quad (15)$$

As mentioned before, when the concentration parameter $\kappa = 0$, the $\mathbb{C}W$ model collapses in the spherical uniform distribution whose density is the constant $(2\pi^k)/(k-1)$. Studying the limit behavior of the proposed entropy measures when the concentration parameter tends to zero is important. The next corollary approaches this question as follows.

Corollary 2 *Let $\mathbf{z} \sim \mathbb{C}W_{k-1}(\boldsymbol{\mu}, \kappa)$. Fixing $\boldsymbol{\mu}$ and $\beta \in \mathbb{R}_+ - \{1\}$ arbitrary then*

$$\lim_{\kappa \rightarrow 0} H_R^\beta(\boldsymbol{\mu}, \kappa) = \lim_{\kappa \rightarrow 0} H_S(\boldsymbol{\mu}, \kappa) = \left[\frac{2\pi^k}{(k-1)!} \right].$$

The proof of all results above is given in Appendix A. Now, we briefly investigate the behavior of the Shannon and Rényi entropies when increasing the concentration parameter.

Consider the Rényi measure order $\beta \in \{0.1, 0.5, 0.9, 2.0, 3.0, 5.0\}$, κ varying from 0 to 10 and the number of landmarks $k+1=5$ (case which will be analyzed in our simulation study). Figure 1a displays the Shannon and Rényi entropy curves. Note that when κ is very close to zero, the values of entropies are similar, as expected from the Corollary 2. Moreover, the behavior of measures indicates more concentrated data tend to have smaller values of entropy. To illustrate the last statement, Fig. 1b shows the plot of three samples drawn from the $\mathbb{C}W_1$ distribution. As discussed by Mardia (1989) and Dryden and Mardia (1998), these data are pre-shapes of triangles, which can be represented either $\mathbb{C}S^1$ or $\mathbb{S}^3$, being the real pre-shape sphere. In this case, the $\mathbb{C}W$ and $\mathbb{C}B$ models are similar. Specifically, we assume they have parameters $(\lambda_1, \lambda_2) = (\kappa, 0)$; i.e., the value of λ_1 given $\lambda_2 = 0$ and $\boldsymbol{\mu} = [0, 1]^\top$ is equal to κ by the identity $-\mathbf{A} = \kappa(\mathbb{I}_2 - \boldsymbol{\mu}\boldsymbol{\mu}^*)$. It is noticeable that the increasing of κ causes the concentration high degree in pre-shape data, implying that small values of entropy

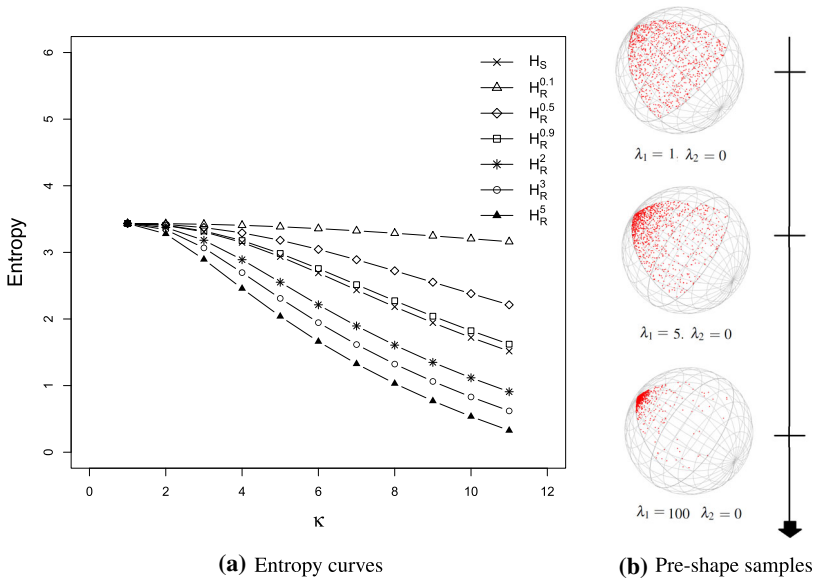


Fig. 1 Illustration for both **a** the values of Shannon and Rényi entropies when $\beta \uparrow 1$ and κ increases, **b** behavior of triangles pre-shapes over the unit sphere

may represent most concentrated spherical phenomena. This conclusion is important because it simplifies the behavior of complex vector data on the unit sphere to a real scale quantity, whose distribution and associated quantiles are more easily studied.

3.2 Pivotal statistics

In this subsection, we present some preliminary results that support the hypothesis tests based on the Rényi and Shannon entropies. These tests are discussed in Salicrú et al. (1994) and Arellano-Valle et al. (2017). They have established KL divergences between generalized skew-normal distributions and normal once in terms of their negentropies in order to develop hypothesis testing for normality. The next Lemma was developed by Pardo (2005, p. 60).

Lemma 1 Let $\hat{\theta}$ be the MLE of θ based on a random sample of size n drawn from a model with pdf $f(\mathbf{x}; \theta)$. Assume the regularity conditions [mentioned in Pardo (2005, pp. 58–59) are valid], $\varphi_{\mathcal{M}} \in C^1([0, \infty))$ (i.e., $\varphi_{\mathcal{M}}$ is derivable continuously in the interval $[0, \infty)$) and there exist a measurable and μ -integrable function $F(x)$ such that

$$\left| \varphi'_{\mathcal{M}}(f(\mathbf{x}; \theta)) \frac{\partial f(\mathbf{x}; \theta)}{\partial \theta_j} \right| < F(\mathbf{x}), \quad j = 1, \dots, k;$$

and

$$\varphi_{\mathcal{M}}(f(\mathbf{x}; \theta)) = \begin{cases} -f(\mathbf{x}; \theta) \log[f(\mathbf{x}; \theta)], & \mathcal{M} = S \\ [f(\mathbf{x}; \theta)]^\beta, & \mathcal{M} = R \text{ and } \beta \in \mathbb{R}_+ - \{1\}. \end{cases}$$

Then

$$\sqrt{n} \left[H_{\mathcal{M}}(\hat{\boldsymbol{\theta}}) - H_{\mathcal{M}}(\boldsymbol{\theta}) \right] \xrightarrow[n \rightarrow \infty]{d} N(0, \sigma_{\mathcal{M}}^2(\boldsymbol{\theta})),$$

where $N(\mu, \sigma^2)$ denotes the normal distribution with mean μ and variance σ^2 , the notation “ $\xrightarrow{d}$ ” represents the convergence in distribution,

$$\sigma_{\mathcal{M}}^2(\boldsymbol{\theta}) = \mathbf{T}_{\mathcal{M}}^{\top} \mathbf{I}_F(\boldsymbol{\theta})^{-1} \mathbf{T}_{\mathcal{M}}, \quad (16)$$

$\mathbf{I}_F(\boldsymbol{\theta})$ is the FIM and $\mathbf{T}_{\mathcal{M}} = \{\mathbf{t}_{\mathcal{M}_1}, \mathbf{t}_{\mathcal{M}_2}, \dots, \mathbf{t}_{\mathcal{M}_k}\}^{\top}$ such that

$$\mathbf{t}_{\mathcal{M}_j} = \frac{\partial H_{\mathcal{M}}(\boldsymbol{\theta})}{\partial \theta_j}, \quad j = \{1, 2, \dots, k\}.$$

Based on the Lemma 1, in order to find the expression for the variance with respect to each considered entropy in this paper, we first determine the components of $\mathbf{T}_{\mathcal{M}}$ in the following proposition.

Proposition 1 Let $\mathbf{z} \sim \mathbb{CB}_{k-1}(\mathbf{A})$ with the Shannon entropy given in (13), we have that the vector $\mathbf{T}_S = (t_{S_1}, t_{S_2}, \dots, t_{S_p})^{\top}$ in (16) is given by

$$t_{S_i} = \lambda_i \left(\frac{1}{c(\Delta)^2} \left(\frac{\partial c(\Delta)}{\partial \lambda_i} \right)^2 - \frac{1}{c(\Delta)} \frac{\partial^2 c(\Delta)}{\partial \lambda_i^2} \right), \quad \text{for } i = 1, \dots, k,$$

where $\Delta = \text{diag}(\lambda_1, \dots, \lambda_{k-1})$ is the eigenvalues matrix of $\mathbf{A}$. With respect to the Rényi entropy in (12), we have the vector $\mathbf{T}_R = \{t_{R_1}, t_{R_2}, \dots, t_{R_p}\}^{\top}$ such that

$$t_{R_i} = \frac{1}{(1 - \beta)} \left[\frac{1}{c(\beta \Delta)} \frac{\partial c(\beta \Delta)}{\partial \lambda_j} - \frac{\beta}{c(\Delta)} \frac{\partial c(\Delta)}{\partial \lambda_i} \right],$$

where $\beta \in \mathbb{R}_+ - \{1\}$.

Let $\mathbf{x}_1, \dots, \mathbf{x}_r$ such that $\mathbf{x}_j \sim \mathbb{CB}_{k-1}(\mathbf{A}_j)$, be populational random vectors $\mathbf{A}_j$ for $j = 1, 2, \dots, r$ and r is the number of under study populations. Consider that we are interested in testing the null hypothesis

$$\mathbb{H}_0 : H_{\mathcal{M}}(\mathbf{A}_1) = H_{\mathcal{M}}(\mathbf{A}_2) = \dots = H_{\mathcal{M}}(\mathbf{A}_r) = D_0$$

versus $\mathbb{H}_1 : \exists j, k \in \{1, \dots, r\}$ such that $H_{\mathcal{M}}(\mathbf{A}_j) \neq H_{\mathcal{M}}(\mathbf{A}_k)$. Let $\hat{\mathbf{A}}_j$ be the ML estimator for $\mathbf{A}_j$ based on a random sample of size n_j drawn from $\mathbf{x}$. It follows from Lemma 1 that

$$\frac{\sqrt{n_j} (H_{\mathcal{M}}(\hat{\mathbf{A}}_j) - D_0)}{\sigma_{\mathcal{M}}(\hat{\mathbf{A}}_j)} \xrightarrow[n \rightarrow \infty]{d} N(0, 1)$$

for $j = 1, 2, \dots, r$, $\mathcal{M} \in \{R, S\}$, and D_0 is an unknown constant. So

$$\sum_{j=1}^r \frac{n_j (H_{\mathcal{M}}(\hat{\mathbf{A}}_j) - D_0)^2}{\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)} \xrightarrow[n_j \rightarrow \infty]{d} \chi_r^2.$$

As D_0 is an unknown constant in practice, Pardo (2005) suggested to rewrite this statistic as

$$\sum_{j=1}^r \frac{n_j (H_{\mathcal{M}}(\hat{\mathbf{A}}_j) - D_0)^2}{\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)} = \sum_{j=1}^r \frac{n_j (H_{\mathcal{M}}(\hat{\mathbf{A}}_j) - \bar{D})^2}{\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)} + \sum_{j=1}^r \frac{n_j (\bar{D} - D_0)^2}{\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)} \quad (17)$$

where

$$\bar{D} = \left[\sum_{j=1}^r \frac{n_j}{\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)} \right]^{-1} \sum_{j=1}^r \frac{n_j H_{\mathcal{M}}(\hat{\mathbf{A}}_j)}{\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)}.$$

Pardo (2005, p. 66) showed that the second parcel of the right side of (17) is distributed as a chi-square distribution with 1 degree of freedom. Thus,

$$S_{\mathcal{M}}(\hat{\mathbf{A}}_1, \hat{\mathbf{A}}_2, \dots, \hat{\mathbf{A}}_r) = \sum_{j=1}^r \frac{n_j (H_{\mathcal{M}}(\hat{\mathbf{A}}_j) - \bar{D})^2}{\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)} \xrightarrow[n \rightarrow \infty]{d} \chi_{r-1}^2. \quad (18)$$

The amount $\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)$ is obtained from the Lemma 1 and Proposition 1. Thus, the following decision rule ensues: the null hypothesis is rejected if

$$S_{\mathcal{M}}(\hat{\mathbf{A}}_1, \hat{\mathbf{A}}_2, \dots, \hat{\mathbf{A}}_r) > \chi_{(r-1), \alpha}^2,$$

where $\chi_{(r-1), \alpha}^2$ denotes the 100%(1 - α) percentile of the chi-squared distribution with $r - 1$ (r is the number of population) degrees of freedom, fixing the significance level at α . A last note about the use of (18) is worth. For majority of experiments under multiple samples, the computation of $\sigma_{\mathcal{M}}^2(\hat{\mathbf{A}}_j)$ may be problematic once this quantity depends of (8). In this case, we suggest to compute the stochastic entropy variance by means of bootstrap, such like it will be discussed in application.

4 Numerical results

Numerical results related to the proposed tests are presented in this section. First, their performance is evaluated for generated $\mathbb{CB}$ data. To this, estimated sizes and powers are considered to compare the proposed methods, next our proposals are applied to real data. All results were computed using the R functions, in particular for the application we use the package *shapes* (for more details, see <https://cran.r-project.org/web/packages/shapes/index.html>).

4.1 Simulation study

Initially, the asymptotic behavior of the proposed pivotal statistics is investigated. A Monte Carlo simulation study was performed with 5000 replicas and, over each one, were computed Shannon and Rényi statistics (with order 0.48, justified subsequently) in (18) for two ($r = 2$) samples with size $n = 50$ and highly concentrated spherical behavior. According to Kent et al. (2004), the matrix parameters $\mathbf{A}$ and $\mathbf{A} + \Upsilon \mathbf{I}_k$, for $\Upsilon \in \mathbb{R}$, define the same distribution. Hence, without loss of generality, we may shift the eigenvalues of $\mathbf{A}$ so that they are non-positive with the largest one equaling 0. Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k = 0$ denote the eigenvalues of $-\mathbf{A}$; i.e., λ_j , here corresponds to $-\lambda_j$ in (4), with $j = 1, \dots, k$. Figures 2a and 3 display theoretical and empirical densities for Rényi and Shannon statistics, respectively, as well as the respective quantile–quantile plots (QQ-plots) in Figs. 2b and 4. It was observable that the Shannon statistic failed (for $n = 50, 100$) to follow the chi-squared distribution; however, for $n = 200$ it looks good. While the Rényi version at the order 0.48 was successful. The Kolmogorov–Smirnov (KS) test (see Darling 1957) was employed to the array of observed $S_R^{0.48}$ statistics, yielding p value 0.2669; i.e., the hypothesis that the observed array comes from the chi-square distribution is not rejected to $\alpha \leq 5\%$. In order to illustrate that the Shannon entropy-based test achieves the chi-squared behavior in some sample size, Figs. 3 and 4 also include the sample sizes $n = 100$ and 200. However, the future discussion focuses on sizes smaller than 100. Although the Shannon test did not work well, it is noticed that there is an order, say β_{otm} , that can optimize the Rényi test in the sense of proximity to the adopted nominal level.

Now, we can do a deeper study to confirm the existence of an order, β_{otm} , for S_R^β , which will allow this statistic to be acceptable on small and moderate samples sizes. We suspect initially that the optimized value depends of the choice on the all quantities involved in tests proposed in Sect. 3: number of landmarks ($k + 1 = 5$), sample sizes (30, 50, 70, 100, 150), number of populations ($r \in \{2, 3, 5, 6\}$) and two

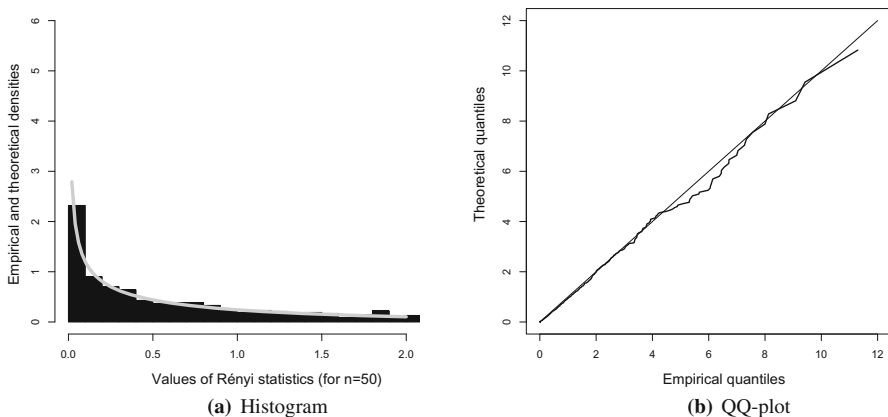


Fig. 2 Histogram and qq-plot of the Rényi statistics under high concentration for samples coming of the $\mathbb{CB}$ model. For simulation procedure we consider: eigenvalues vector $\lambda = (99, 42, 1, 0)$, sample sizes $n_1 = n_2 = n = 50$, populacional number $r = 2$ and $k + 1 = 5$ landmarks

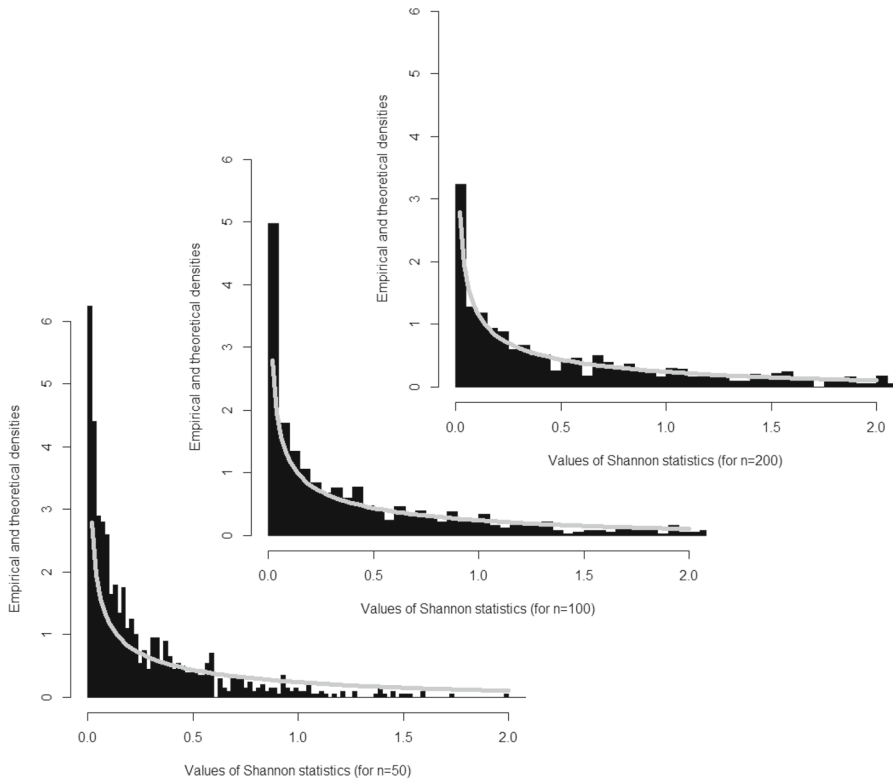


Fig. 3 Histogram of the Shannon statistics under high concentration for samples coming of the $\mathbb{CB}$ model. For simulation procedure we consider: eigenvalues vector $\lambda = (99, 42, 1, 0)$, sample sizes $n_1 = n_2 = n = 50, 100, 200$, populacional number $r = 2$ and $k + 1 = 5$ landmarks

concentration degrees (low and high). The performance of S_S and S_R^β (with order parameter $\beta \in \{0.5, 0.9, \beta_{\text{otm}}\}$) tests was measured under 5000 Monte Carlo replicas.

Tables 1 and 2 present values for the test size under the low and high concentration scenarios. For high concentration, it was observed some cases on which the non-optimized S_R^β test have presented good results: $r = 2, 3, \beta \in \{0.5, 0.9\}$ and small size. However, in general, the test based in the Shannon entropy together with those in terms of the Rényi measure with order 0.5 and 0.9 have shown unsatisfactory results. But, for both concentration scenarios, we have found a way to optimize β so that the rejection rates under the null hypothesis have achieved a good result. We use the methodology described on Algorithm 1 to determine an acceptable $\beta, \beta_{\text{otm}}$.

In order to illustrate the discriminating capability of the Rényi test with the optimized order parameter comparatively to the remainder ones, we have quantified their power functions for some cases. Figure 5a, b shows the power function performance for low concentrations for sample sizes $n = 70$ and 100. It was considered the landmarks number $k + 1 = 5$, 1000 Monte Carlo replicas, the sample size $n = \{70, 100\}$ and population number $r = 2$. In the simulation procedure, we take the eigenvalue vector $\lambda = (5, 2, 1, 0)$ to define the null hypothesis to tests: S_R^β with $\beta = (0.5, 0.9, 1.3, 1.6)$

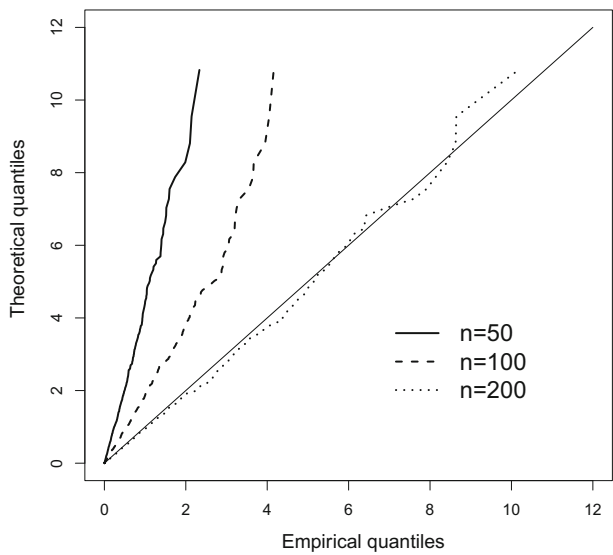


Fig. 4 Q-qplot of the Shannon statistics under high concentration for samples coming of the $\mathbb{C}B$ model. For simulation procedure we consider: eigenvalues vector $\lambda = (99, 42, 1, 0)$, sample sizes $n_1 = n_2 = n = 50, 100, 200$, populacional number $r = 2$ and $k + 1 = 5$ landmarks

Table 1 Rejection rates under $\mathbb{H}_0$ of entropy-based tests for low concentration scenario

λ	r	n	S_S	$S_R^{0.5}$	$S_R^{0.9}$	$S_R^{\beta_{otm}}$	β_{otm}
(5,2,1,0)	2	30	0.2244	0.0022	0.0066	0.0492	1.9
		50	0.3704	0.0024	0.0084	0.046	1.57
		70	0.4372	0.0016	0.0072	0.0488	1.6
		100	0.5692	0.017	0.0352	0.048	1.3
	3	30	0.243	0.0028	0.0078	0.0474	1.8
		50	0.4138	0.0014	0.0034	0.0496	1.7
		70	0.5014	0.0022	0.0066	0.0464	1.65
		100	0.6184	0.0206	0.0364	0.0476	1.25
	5	30	0.9866	0.9156	0.9134	0.0506	0.028
		50	0.995	0.9814	0.9792	0.0476	0.041
		70	0.9944	0.9838	0.986	0.0452	0.048
		100	0.9858	0.9722	0.9764	0.051	0.05
	6	30	0.9886	0.9262	0.924	0.0426	0.025
		50	0.9964	0.9764	0.9748	0.051	0.042
		70	0.9962	0.9858	0.984	0.0506	0.048
		100	0.9858	0.9736	0.9726	0.0512	0.049

The adopted nominal level $\alpha = 5\%$

Table 2 Rejection rates under H_0 of entropy-based tests for high concentration scenario

λ	r	n	S_S	$S_R^{0.5}$	$S_R^{0.9}$	$S_R^{\beta_{\text{otm}}}$	β_{otm}
(99,42,1,0)	2	30	0	0.015	0.0064	0.0478	0.18
		50	0	0.0482	0.0318	0.0498	0.48
		70	2e−04	0.0918	0.0704	0.0608	1.68
		100	0.001	0.1642	0.1308	0.125	1.22
	3	30	0	0.0036	0.0016	0.0448	0.12
		50	0	0.029	0.0186	0.0498	0.35
		70	0	0.0768	0.0538	0.0524	0.95
		100	2e−04	0.1454	0.1056	0.0468	1.36
	5	30	0	0.2556	0.0074	0.0462	4.4
		50	0	0.7354	0.081	0.0554	2.3
		70	4e−04	0.9448	0.2404	0.0496	1.8
		100	0.0084	0.9958	0.4992	0.0492	1.55
	6	30	0	0.2676	0.008	0.0456	0.71
		50	0	0.7676	0.0848	0.0462	0.99
		70	2e−04	0.9568	0.2506	0.0488	1.15
		100	0.008	0.9968	0.5244	0.053	1.6

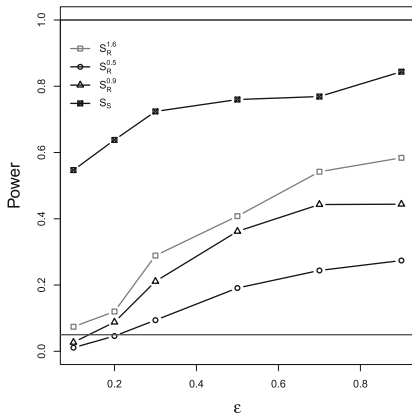
The adopted nominal level $\alpha = 5\%$

Algorithm 1 Procedure to obtain β_{otm} with minimizes the rejection rates under the null hypothesis

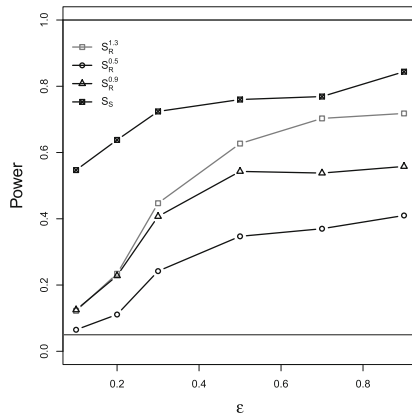
- 1: Given r samples supposed to come from the same population having the support over the unit sphere, obtain B sets with r vectors of size n .
- 2: Define a grid of values for β in S_R^β as wide as possible and after determine an array with the rejection rates in terms of values of the adopted grid.
- 3: Choose β_{otm} as the value of β which minimizes the rejection rates in the array of the previous item.

($\beta_{\text{otm}} = 1.3$ for $n = 100$ and $\beta_{\text{otm}} = 1.6$ for $n = 70$, according to Table 1) and S_S . From this, we generate the first sample with λ and the second sample with $\underline{\lambda} = (5 \cdot (1 - \epsilon)^{2.5}, 2 \cdot (1 - \epsilon)^{2.5}, 1, 0)$ where $\epsilon = (0.10, 0.20, 0.30, 0.50, 0.70, 0.90)$. Note that the Shannon test provided a power bigger than the Rényi test. However, as discussed, the S_S empirical test size showed unacceptable before considered sample sizes. On the other hand, among the possible Rényi tests, the $S_R^{\beta_{\text{otm}}}$ power outperformed ones due to the remainder tests. Further, the Rényi test power increases with the growth of the sample size.

Figure 6 displays the power function for high concentration scenario. For this, we consider $\lambda = (99, 42, 1, 0)$ to specify the null hypothesis of tests: S_R^β with $\beta = (0.5, 0.9, 1.68)$ and S_S . The choose $\beta = 1.68$ is aligned with $n = 70$ in Table 2. Here, we consider only one sample size for the power assessment on high concentration because tests performed accurately at $n = 70$. To that end, we generated the first sample with λ and the second with $\underline{\lambda} = (99 \cdot (1 - \epsilon), 42 \cdot (1 - \epsilon), 1)$ such that $\epsilon = [0.1, 0.2, 0.3, 0.4]^\top$. We note the performance of the Rényi test power was better



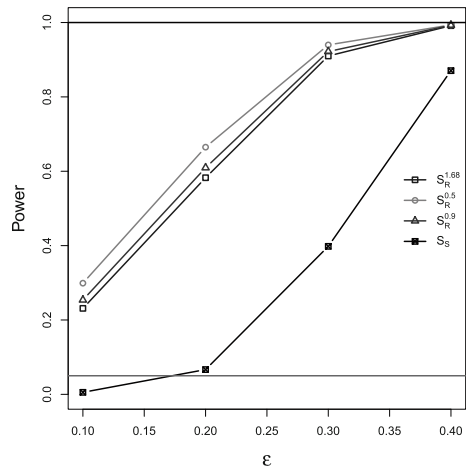
(a) Power function under low concentration with sample size $n_1 = n_2 = 70$.



(b) Power function under low concentration with sample size $n_1 = n_2 = 100$.

Fig. 5 Empirical power function under low concentration. For this numeric procedure we consider: eigenvalue vector $\lambda = (5, 2, 1, 0)$, order parameter $\beta \in \{0.5, 0.9, 1.3, 1.6\}$ for the Rényi statistic, populacional number $r = 2$ and $k + 1 = 5$ landmarks

Fig. 6 Empirical power function under high concentration. For this numeric procedure we consider: eigenvalue vector $\lambda = (99, 42, 1, 0)$, order parameter $\beta \in \{0.5, 0.9, 1.6\}$ for the Rényi statistic, sample size $n_1 = n_2 = n = 70$ populacional number $r = 2$ and $k + 1 = 5$ landmarks

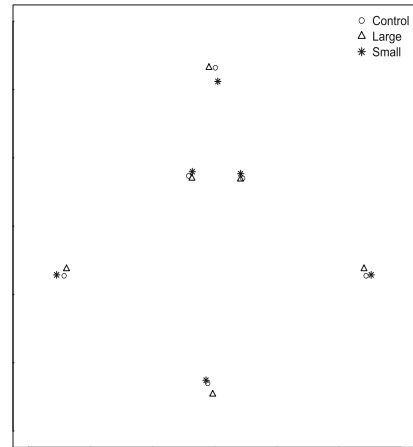


than for the Shannon test. In this case, the empirical power of the $S_R^{\beta_{\text{otm}}}$ test was the smallest in the Rényi class, but all these tests performed very well and almost similarly.

4.2 Application to real data

We employ the derived tests to identify patterns over a database of measurements on the second thoracic vertebra T2 of mice, described by Dryden and Mardia (2016). We wish to quantify how the body weight influences the shape of the mouse T2 vertebrae. We work with three groups of mice: control (with 30 elements), large (23) and small (23). The control group contains unselected mice, while the large and

Fig. 7 Illustration of the second thoracic vertebra T2 of the group mice. **a** A vertebra T2 from mouse which were registered using the mean shape of full Procrustes fit with six landmark



small groups are formed by mice which were selected having large and small body weight, respectively.

Figure 7a presents the full Procrustes fit [see Goodall (1991) and Dryden and Mardia (2016)] of landmarks data concerning with the three groups of mice. For each group, we obtain the mean shapes of the fitted data and plot the $k + 1 = 6$ landmarks referring to the three groups (control, large and small). This graph was made using the function `procrustes2d` on the package shapes of the software R.

Before real data, we have verified the FIM in (8) was singular and, consequently, the use of (16) was not possible. To overcome this issue, we obtained $\sigma_{\mathcal{M}}^2$ in (16) by means of the nonparametric bootstrap methodology. Initially, in order to construct a random framework from a real dataset, we performed 750 resampling with replacement over each considered group. Since the implementation had some problems, an alternative method was used to compute the estimated asymptotic variance. After, we estimated the variance ($\sigma_{\mathcal{M}}^2$) using the following expression

$$\hat{\sigma}_{\mathcal{M}}^2 = \frac{1}{750} \sum_{j=1}^{750} \left(H_{\mathcal{M}}^{(j)} - \overline{H_{\mathcal{M}}} \right)^2,$$

where $H_{\mathcal{M}}^{(j)}$ is the entropy $\mathcal{M} = \{S, R\}$ which was estimated over the j th bootstrap sample and $\overline{H_{\mathcal{M}}}$ is the bootstrap average of estimated entropies.

Table 3 shows values for high concentration scenario for pivotal statistics based on the Rényi and Shannon entropies, but only the test based on the Rényi entropy rejected the null hypothesis adopting $\alpha = 5\%$, independent of the considered samples. On the other hand, the test based on the Shannon entropy never rejects $\mathbb{H}_0$. Regarding the difference in groups of entropies on the set $\{\text{small}, \text{large}, \text{control}\}$, it is expected there are differences in pairs (small, control) and (large, control). Thus, these results seem to recommend in practice the Rényi test may be more adequate to detect the variability in planar pre-shape over the complex unitary sphere.

Table 3 Rejection rates under H_0 , values of proposed statistics and their variances based on real data for high concentration

Tests	$\{n_1, n_2, n_3\}$	$\bar{S}_{\mathcal{M}}$	p values	$\hat{\sigma}_{\mathcal{M}}^2(\hat{A}_1)$	$\hat{\sigma}_{\mathcal{M}}^2(\hat{A}_2)$	$\hat{\sigma}_{\mathcal{M}}^2(\hat{A}_3)$
S_S	{10, 10, 10}	0.06285198	0.9690627	0.2868608	0.1290481	0.2586201
$S_R^{0.00775}$		6.233742	0.04429555	0.1017824	0.06037482	0.07393719
S_S	{15, 15, 15}	0.1191108	0.9421833	0.1383523	0.0916418	0.1015166
$S_R^{0.00746}$		6.057827	0.04836815	0.05435652	0.04583011	0.04742249
S_S	{30, 23, 23}	3.543456	0.1700389	0.02320577	0.06870133	0.1096711
$S_R^{0.0068}$		11.80529	0.00273221	0.01091856	0.02620905	0.04358041

The adopted nominal level $\alpha = 5\%$

5 Conclusions

This paper has presented two entropy-based r -samples hypothesis tests for checking common variability in planar-shape data. The complex Bingham ($\mathbb{CB}$) model is assumed for the data. We have provided expressions for the $\mathbb{CB}$ Shannon and Rényi entropies. Some properties of these measures were also explored; such as new expressions for the complex Watson ($\mathbb{CW}$) particular model and limit relations between those entropies. Geometrical essays have indicated that the interpretation of entropy as a variability measure is aligned with the degree of concentration of planar pre-shape over the unit sphere. Finally, entropy-based hypothesis tests for variability in planar shape were proposed as well.

The performance of our proposals has been quantified by means of a Monte Carlo simulation study. The Rényi entropy test with optimized order parameter obtained the best empirical test size in all the cases. Among possible tests to be defined from the Rényi class, that last achieved either lesser (but closed to) empirical power than remainder in the high concentration scenarios or expressively higher for low concentration. Finally, we illustrated the potentiality of these measures over real data to evaluate the effect of body weight over the shape of mouse vertebrae. The Rényi (in particular, with β optimized order) test got better performance than one in terms of the Shannon entropy. We also pointed out in practice a bootstrap based strategy to outperform the computational instabilities to obtain variances of entropies before singularity evidence in the Fisher information matrix. We also pointed out in practice a bootstrap based strategy to outperform the computational instabilities to obtain variances of entropies before singularity evidence in the Fisher information matrix.

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A Proofs related to the Shannon and Rényi entropies

Proof of the Theorem 1 Let $\mathbf{z} \sim \mathbb{C}\mathcal{B}_{k-1}(\mathbf{A})$ be a random variable with pdf $f(\mathbf{z}; \mathbf{A})$, so the Shannon entropy is given by

$$\begin{aligned} H_S(\mathbf{A}) &= \mathbb{E}(-\log f(\mathbf{z}; \mathbf{A})) = \mathbb{E}\{-\log[c^{-1}(\mathbf{A}) \exp(\mathbf{z}^* \mathbf{A} \mathbf{z})]\} \\ &= \mathbb{E}[\log c(\mathbf{A}) - \mathbf{z}^* \mathbf{A} \mathbf{z}] = \log c(\mathbf{A}) - \mathbb{E}(\mathbf{z}^* \mathbf{A} \mathbf{z}). \end{aligned}$$

To solve the expectation involving the quadratic forms, we use a variable change based on the Kent's polar shape coordinates. Kent (1994) proposed some nonstandard polar coordinates on the pre-shape sphere. Given a point $(z_1, z_2, \dots, z_k)^\top$ on $\mathbb{C}\mathcal{S}^{k-1}$, we transform it to $(s_1, \dots, s_{k-1}, \theta_1, \dots, \theta_k)$, where

$$\operatorname{Re}(z_j) = s_j^{1/2} \cos(\theta_j) \text{ and } \operatorname{Im}(z_j) = s_j^{1/2} \sin(\theta_j),$$

for $j = 1, \dots, k$, $s_j \geq 0$, $\theta \in [0, 2\pi)$ and $s_k = 1 - \sum_{i=1}^{k-1} s_i$. The coordinates $s_1, s_2, \dots, s_{k-1}$ are on the $(k-1)$ -dimensional unit simplex in $\mathbb{R}^{k-1}$ defined by

$$\mathcal{S}_{k-1} = \left\{ (s_1, \dots, s_{k-1})^\top : s_j \geq 0 \text{ and } \sum_{j=1}^{k-1} s_j \leq 1 \right\}.$$

The vector $\mathbf{s} = (s_1, \dots, s_{k-1})^\top$ follows the multivariate exponential distribution truncated to a simplex with density given by (Kent et al. 2004)

$$f_{TE}(\mathbf{s}) = b(\lambda) \prod_{j=1}^{k-1} \lambda_j \exp(-\lambda_j s_j), \quad s_j \geq 0, \quad j = 1, \dots, k-1,$$

where $b(\lambda) = \left\{ \prod_{j=1}^{k-1} [1 - \exp(-\lambda_j)] \right\}^{-1}$. Then,

$$\begin{aligned} \mathbf{z}^* \mathbf{A} \mathbf{z} &= \operatorname{tr}(\mathbf{z}^* \mathbf{A} \mathbf{z}) = \operatorname{tr} \left(\sum_{j=1}^k \lambda_j \gamma_j \gamma_j^* \mathbf{z}_j^* \mathbf{z}_j \right) \\ &= \operatorname{tr} \left(\sum_{j=1}^k \lambda_j \gamma_j^* \gamma_j \mathbf{z}_j^* \mathbf{z}_j \right) = \sum_{j=1}^k \lambda_j \mathbf{z}_j^* \mathbf{z}_j = \sum_{j=1}^k \lambda_j s_j, \end{aligned}$$

Thus,

$$\mathbb{E}(\mathbf{z}^* \mathbf{A} \mathbf{z}) = \sum_{j=1}^k \lambda_j \mathbb{E}(s_j).$$

According to Kent (1994), we have that

$$\mathbb{E}(s_j) = \frac{\partial \log c(\mathbf{\Delta})}{\partial \lambda_j} = \frac{1}{c(\mathbf{\Delta})} \frac{\partial c(\mathbf{\Delta})}{\partial \lambda_j},$$

where the first derivative of normalizing constant is given by

$$\frac{\partial c(\boldsymbol{\Delta})}{\partial \lambda_j} = 2\pi^k \left[\sum_{l \neq j}^k \frac{a_l \exp(\lambda_l)}{(\lambda_l - \lambda_j)} + a_j \exp(\lambda_j) - a_j b_j \exp(\lambda_j) \right].$$

Using the change above transformations, we have to

$$\begin{aligned} H_S(\mathbf{A}) &= \log c(\mathbf{A}) - \mathbb{E}(\mathbf{z}^* \mathbf{A} \mathbf{z}) = \log c(\mathbf{A}) - \mathbb{E} \left(\sum_{i=1}^k \lambda_i s_i \right) \\ &= \log c(\mathbf{A}) - \sum_{i=1}^k \lambda_i \mathbb{E}(s_i) \\ &= \log c(\mathbf{A}) - \sum_{i=1}^k \lambda_i \frac{1}{c(\boldsymbol{\Delta})} \frac{\partial c(\boldsymbol{\Delta})}{\partial \lambda_i} = \log c(\boldsymbol{\Delta}) - \sum_{i=1}^k \lambda_i \frac{1}{c(\boldsymbol{\Delta})} \frac{\partial c(\boldsymbol{\Delta})}{\partial \lambda_i}. \end{aligned}$$

Therefore,

$$H_S(\mathbf{A}) = \log c(\boldsymbol{\Delta}) - \sum_{i=1}^k \lambda_i \frac{1}{c(\boldsymbol{\Delta})} \frac{\partial c(\boldsymbol{\Delta})}{\partial \lambda_i}.$$

The Rényi entropy with order parameter $\beta \in \mathbb{R}_+ - \{1\}$ is given by

$$\begin{aligned} H_R^\beta(\mathbf{A}) &= \frac{1}{(1-\beta)} \log \{ \mathbb{E}[f^{\beta-1}(\mathbf{z}; \mathbf{A})] \} \\ &= \frac{1}{(1-\beta)} \log \left\{ \int_{\mathbb{C}S^{k-1}} [c^{-1}(\mathbf{A}) \exp(\mathbf{z}^* \mathbf{A} \mathbf{z})]^\beta d\mathbf{z} \right\} \\ &= \frac{1}{(1-\beta)} \log \left\{ \int_{\mathbb{C}S^{k-1}} c^{-\beta}(\mathbf{A}) \exp(\mathbf{z}^* \beta \mathbf{A} \mathbf{z}) d\mathbf{z} \right\} \\ &= \frac{1}{(1-\beta)} \log \left\{ c^{-\beta}(\mathbf{A}) c(\beta \mathbf{A}) \int_{\mathbb{C}S^{k-1}} c^{-1}(\beta \mathbf{A}) \exp(\mathbf{z}^* \beta \mathbf{A} \mathbf{z}) d\mathbf{z} \right\} \\ &= \frac{1}{(1-\beta)} \log \frac{c(\beta \mathbf{A})}{c^\beta(\mathbf{A})} = \frac{1}{(1-\beta)} [\log c(\beta \mathbf{A}) - \beta \log c(\mathbf{A})]. \end{aligned}$$

Therefore, the Rényi entropy is

$$H_R^\beta(\mathbf{A}) = \frac{1}{(1-\beta)} [\log c(\beta \mathbf{A}) - \beta \log c(\mathbf{A})].$$

□

The Lebesgue dominated convergence (Billingsley 2008) will be used to obtain the Shannon entropy in Theorem 2.

Theorem 2 (Lebesgue dominated convergence theorem (TCL)) *If $|f_n| \leq g$ almost everywhere, where g is integrable, and if $f_n \rightarrow f$ almost everywhere, then f and f_n are integrable and $\int f_n d\eta \rightarrow \int f d\eta$.*

Proof of the Theorem 1 (a) The Shannon entropy is obtained as

$$H_S(\boldsymbol{\mu}, \kappa) = \mathbb{E}_{\mathbf{z}} \left\{ -\log \left[c_1^{-1}(\kappa) \exp \left(\kappa |\mathbf{z}^* \boldsymbol{\mu}|^2 \right) \right] \right\} = \log [c_1(\kappa)] - \kappa \mathbb{E}_{\mathbf{z}} \left(|\mathbf{z}^* \boldsymbol{\mu}|^2 \right). \quad (19)$$

To find a closed-form expression of $\mathbb{E}_{\mathbf{z}}(|\mathbf{z}^* \boldsymbol{\mu}|^2)$, we resort the transform Kent's polar coordinates on $\mathbb{C}\mathcal{S}^{k-1}$ given in the proof of the Theorem 1. Initially, we take $\boldsymbol{\mu} = (1, 0, \dots, 0)^T$ and after a suitable rotation $|\mathbf{z}^* \boldsymbol{\mu}|^2 = s_1$ (see Mardia Mardia and Dryden 1999, p. 915), this change of variables (in terms of $s_j \in \mathcal{S}_{k-1}$ and $\theta \in [0, 2\pi)$) allows that the complex pre-shape sphere can be identified by $\mathcal{S}_{k-1} \times [0, 2\pi)$ such that the volume measure on $\mathbb{C}\mathcal{S}^{k-1}$ is given by

$$2^{(1-k)} ds_1, \dots, ds_{k-1} d\theta_1, \dots, d\theta_k$$

in this way we have

$$\begin{aligned} \mathbb{E}_{\mathbf{z}}(|\mathbf{z}^* \boldsymbol{\mu}|^2) &= \int_{\mathbb{C}\mathcal{S}^{k-1}} |\mathbf{z}^* \boldsymbol{\mu}|^2 c_1^{-1}(\kappa) \exp(\kappa |\mathbf{z}^* \boldsymbol{\mu}|^2) d\mathbf{z} \\ &= \int_{S_{k-1} \times [0, 2\pi)^k} s_1 c_1^{-1}(\kappa) \exp(\kappa s_1) 2^{(1-k)} ds_1, \dots, ds_{k-1} d\theta_1, \dots, d\theta_k \\ &= c_1^{-1}(\kappa) \int_{S_{k-1} \times [0, 2\pi)^k} s_1 \exp(\kappa s_1) 2^{(1-k)} ds_1, \dots, ds_{k-1} d\theta_1, \dots, d\theta_k \\ &= c_1^{-1}(\kappa) \int_{S_{k-1} \times [0, 2\pi)^k} \frac{\partial}{\partial \kappa} [\exp(\kappa s_1)] 2^{(1-k)} ds_1, \dots, ds_{k-1} d\theta_1, \dots, d\theta_k \\ &\quad \text{using the Theorem 2} \\ &= c_1^{-1}(\kappa) \frac{\partial}{\partial \kappa} \left\{ 2^{(1-k)} \int_{S_{k-1}} \int_{[0, 2\pi)^k} [\exp(\kappa s_1)] ds_1, \dots, ds_{k-1} d\theta_1, \dots, d\theta_k \right\} \\ &= c_1^{-1}(\kappa) \frac{\partial}{\partial \kappa} [c_1(\kappa)]. \end{aligned}$$

According to Abramowitz and Stegun (1964, p. 507), the n th derivative of confluent hypergeometric function is given by

$$\frac{d^n}{dx^n} {}_1F_1(a; b; x) = \frac{(a)_n}{(b)_n} {}_1F_1(a; b; x),$$

where $(a)_n = a(a+1)(a+2) \dots (a+n-1)$, $(a)_0 = 1$. Then,

$$\begin{aligned} \mathbb{E}_{\mathbf{z}}(|\mathbf{z}^* \boldsymbol{\mu}|^2) &= c_1^{-1}(\kappa) \frac{\partial}{\partial \kappa} [c_1(\kappa)] = \left[\frac{(k-1)!}{2\pi^k} \cdot \frac{1}{{}_1F_1(1; k; \kappa)} \right] \\ &\quad \cdot \left[\frac{2\pi^k}{(k-1)!} \frac{1}{k} {}_1F_1(2; k+1; \kappa) \right] \end{aligned}$$

$$= \frac{1}{k} \frac{{}_1F_1(2; k+1; \kappa)}{{}_1F_1(1; k; \kappa)}.$$

To generalize the result to $\boldsymbol{\mu} \neq (1, 0, \dots, 0)^\top$ arbitrary, consider a rotation matrix $\mathbf{R}_{k \times k}$ unitary whose first column is $\boldsymbol{\mu}$. A suitable matrix $\mathbf{R}$ can be obtained using the QR decomposition (see Seber 2008, p. 340; Golub and Van Loan 2012, pp. 223–236; Kurz and Hanebeck 2015; Kurz et al. 2016). Thus, we have $\boldsymbol{\mu} = \mathbf{R}(1, 0, \dots, 0)^\top$ and $\mathbf{R}^* \boldsymbol{\mu} = (1, 0, \dots, 0)^\top$. In the integration procedure, we use the substitution $\mathbf{z} = \mathbf{R}\mathbf{t}$ with $\mathbf{t} \in \mathbb{C}S^{k-1}$. Then, we reduce the problem to the case of $\boldsymbol{\mu} = (1, 0, \dots, 0)^\top$. Thus,

$$\begin{aligned} \mathbb{E}_{\mathbf{z}}(|\mathbf{z}^* \boldsymbol{\mu}|^2) &= \int_{\mathbb{C}S^{k-1}} |\mathbf{z}^* \boldsymbol{\mu}|^2 c_1^{-1}(\kappa) \exp(\kappa |\mathbf{z}^* \boldsymbol{\mu}|^2) d\mathbf{t} \\ &= c_1^{-1}(\kappa) \int_{\mathbb{C}S^{k-1}} |(\mathbf{R}\mathbf{t})^* \boldsymbol{\mu}|^2 \exp(\kappa |(\mathbf{R}\mathbf{t})^* \boldsymbol{\mu}|^2) d\mathbf{t} \\ &= c_1^{-1}(\kappa) \int_{\mathbb{C}S^{k-1}} |\mathbf{t}^* \mathbf{R}^* \boldsymbol{\mu}|^2 \exp(\kappa |\mathbf{t}^* \mathbf{R}^* \boldsymbol{\mu}|^2) d\mathbf{t} \\ &= c_1^{-1}(\kappa) \int_{\mathbb{C}S^{k-1}} |\mathbf{t}^* \mathbf{R}^* \boldsymbol{\mu}|^2 \exp(\kappa |\mathbf{t}^* \mathbf{R}^* \boldsymbol{\mu}|^2) d\mathbf{t} \\ &= c_1^{-1}(\kappa) \int_{\mathbb{C}S^{k-1}} |\mathbf{t}^* (1, 0, \dots, 0)^\top|^2 \exp(\kappa |\mathbf{t}^* (1, 0, \dots, 0)^\top|^2) d\mathbf{t} \\ &= c_1^{-1}(\kappa) \int_{\mathbb{C}S^{k-1}} s_1 \exp(\kappa s_1) 2^{(1-k)} ds_1, \dots, ds_{k-1} d\theta_1, \dots, d\theta_k \\ &= c_1^{-1}(\kappa) \frac{\partial}{\partial \kappa} [c_1(\kappa)] = \frac{1}{k} \frac{{}_1F_1(2; k+1; \kappa)}{{}_1F_1(1; k; \kappa)}. \end{aligned} \quad (20)$$

Therefore, applying (20) in (19), we have $H_S(\boldsymbol{\mu}, \kappa) = \log[c_1(\kappa)] - \frac{\kappa}{k} \frac{{}_1F_1(2; k+1; \kappa)}{{}_1F_1(1; k; \kappa)}$.

(b) The Rényi entropy is given by

$$\begin{aligned} H_R^\beta(\boldsymbol{\mu}, \kappa) &= \frac{1}{(1-\beta)} \log \mathbb{E}_{\mathbf{z}} \left\{ \left[c_1^{-1}(\kappa) \exp(\kappa |\mathbf{z}^* \boldsymbol{\mu}|^2) \right]^{(\beta-1)} \right\} \\ &= \frac{1}{(1-\beta)} \{ (1-\beta) \log c_1(\kappa) + \log \mathbb{E}_{\mathbf{z}} [\exp(\kappa (\beta-1) |\mathbf{z}^* \boldsymbol{\mu}|^2)] \}. \end{aligned} \quad (21)$$

We need to find the closed-form expression for expectation to the right side in (21). Thus,

$$\begin{aligned} &\mathbb{E}_{\mathbf{z}}[\exp(\kappa (\beta-1) |\mathbf{z}^* \boldsymbol{\mu}|^2)] \\ &= \int_{\mathbb{C}S^{k-1}} \exp[\kappa (\beta-1) |\mathbf{z}^* \boldsymbol{\mu}|^2] c_1^{-1}(\kappa) \exp(\kappa |\mathbf{z}^* \boldsymbol{\mu}|^2) d\mathbf{z} \end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{C}S^{k-1}} c_1^{-1}(\kappa) \exp[\kappa(\beta-1)|\mathbf{z}^*\boldsymbol{\mu}|^2 + \kappa|\mathbf{z}^*\boldsymbol{\mu}|^2] d\mathbf{z} \\
&= \frac{c_1(\kappa\beta)}{c_1(\kappa)} \int_{\mathbb{C}S^{k-1}} c_1^{-1}(\kappa\beta) \exp[\kappa\beta|\mathbf{z}^*\boldsymbol{\mu}|^2] d\mathbf{z} = \frac{c_1(\kappa\beta)}{c_1(\kappa)}.
\end{aligned}$$

Then,

$$\begin{aligned}
\mathbf{H}_R^\beta(\boldsymbol{\mu}, \kappa) &= \frac{1}{(1-\beta)} \left\{ (1-\beta) \log c_1(\kappa) + \log \left[\frac{c_1(\kappa\beta)}{c_1(\kappa)} \right] \right\} \\
&= \frac{1}{(1-\beta)} \{ (1-\beta) \log[c_1(\kappa)] + \log[c_1(\kappa\beta)] - \log[c_1(\kappa)] \} \\
&= \frac{1}{(1-\beta)} \{ \log[c_1(\kappa\beta)] - \beta \log[c_1(\kappa)] \}.
\end{aligned}$$

Concluding the proof of the Theorem. $\square$

Proof of the Corollary 2 Considering the parameter $\boldsymbol{\mu}$ fixed and $\beta \in \mathbb{R}_+ - \{1\}$ arbitrary. Making κ tend to zero, we obtain

$$\begin{aligned}
\lim_{\kappa \rightarrow 0} \mathbf{H}_R^\beta(\boldsymbol{\mu}, \kappa) &= \lim_{\kappa \rightarrow 0} \frac{1}{(1-\beta)} \{ \log[c_1(\beta\kappa)] - \beta \log[c_1(\kappa)] \} \\
&= \frac{1}{(1-\beta)} \lim_{\kappa \rightarrow 0} \left\{ \log \left[\frac{2\pi^k}{(k-1)!} {}_1F_1(1; k; \beta\kappa) \right] - \beta \log \left[\frac{2\pi^k}{(k-1)!} \right] \right\} \\
&= \frac{1}{(1-\beta)} \left\{ \log \left[\frac{2\pi^k}{(k-1)!} \right] - \beta \log \left[\frac{2\pi^k}{(k-1)!} \right] \right\} \\
&= \log \left[\frac{2\pi^k}{(k-1)!} \right],
\end{aligned}$$

and Shannon entropy

$$\lim_{\kappa \rightarrow 0} \mathbf{H}_S(\boldsymbol{\mu}, \kappa) = \lim_{\kappa \rightarrow 0} \left\{ \log c_1(\kappa) - \frac{\kappa}{k} \cdot \frac{{}_1F_1(2; k+1; \kappa)}{{}_1F_1(1; k; \kappa)} \right\} = \log \frac{2\pi^k}{(k-1)!}.$$

Proof of the Proposition 1 For the $\mathbb{C}B$ model, the entries of matrices T_R and T_S , respectively, are given by

$$\begin{aligned}
t_{R_i} &= \frac{\partial \mathbf{H}_R(\mathbf{A})}{\partial \lambda_i} = \frac{\partial}{\partial \lambda_i} \left[\frac{1}{(1-\beta)} (\log c(\beta\mathbf{A}) - \beta \log c(\mathbf{A})) \right] \\
&= \frac{1}{(1-\beta)} \left[\frac{\partial \log c(\beta\mathbf{A})}{\partial \lambda_i} - \beta \frac{\partial \log c(\mathbf{A})}{\partial \lambda_i} \right] \\
&= \frac{1}{(1-\beta)} \left[\frac{1}{c(\beta\mathbf{A})} \frac{\partial c(\beta\mathbf{A})}{\partial \lambda_i} - \frac{\beta}{c(\mathbf{A})} \frac{\partial c(\mathbf{A})}{\partial \lambda_i} \right]
\end{aligned}$$

$$= \frac{1}{(1 - \beta)} \left[\frac{1}{c(\beta \mathbf{\Delta})} \frac{\partial c(\beta \mathbf{\Delta})}{\partial \lambda_i} - \frac{\beta}{c(\mathbf{\Delta})} \frac{\partial c(\mathbf{\Delta})}{\partial \lambda_i} \right]$$

and

$$\begin{aligned} t_{S_i} &= \frac{\partial H_S(\mathbf{\Delta})}{\partial \lambda_i} = \frac{1}{c(\mathbf{\Delta})} \frac{\partial c(\mathbf{\Delta})}{\partial \lambda_i} \\ &\quad - \left[\frac{1}{c(\mathbf{\Delta})} \frac{\partial c(\mathbf{\Delta})}{\partial \lambda_i} + \lambda_i \left(-\frac{1}{c(\mathbf{\Delta})^2} \left(\frac{\partial c(\mathbf{\Delta})}{\partial \lambda_i} \right)^2 + \frac{1}{c(\mathbf{\Delta})} \frac{\partial^2 c(\mathbf{\Delta})}{\partial \lambda_i^2} \right) \right] \\ &= \lambda_i \left(\frac{1}{c(\mathbf{\Delta})^2} \left(\frac{\partial c(\mathbf{\Delta})}{\partial \lambda_i} \right)^2 - \frac{1}{c(\mathbf{\Delta})} \frac{\partial^2 c(\mathbf{\Delta})}{\partial \lambda_i^2} \right). \end{aligned}$$

□

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